

Exploring Commenting Behavior in the COVID-19 Super-Topic on Weibo

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ABSTRACT

During the COVID-19 pandemic, Chinese social media platform Weibo created a special super-topic for online users to seek help through posts or provide help through comments. Prior work has analyzed online help-seeking and help-providing messages during crisis situations, however, it is not clear why and how people commented on help-seeking posts to provide support, especially in such a crisis support community. In this study, we interviewed 23 commenters in the COVID-19 super-topic to examine their motivations and strategies for commenting. Our findings showed that geographical proximity and expertise level affected users' commenting behaviors. For example, non-locals or people lacking professional knowledge applied a "copy-and-paste" strategy and acted as a bridge between help-seekers and influencers on social media, while locals and experts focused on detailed information and provided targeted help. Some commenters also exhibited avoidance behaviors due to their sense of powerlessness and frustration in such a long-lasting pandemic. We discuss the implications of our findings for designing social media platforms to better support people in need during crisis situations.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

Help-Providing, Crisis Informatics, COVID-19, Weibo, Super-Topic

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1 INTRODUCTION

Help-seeking behaviors are common on social media sites in crisis situations [4, 32]. People turn to social media to understand the public information space and figure out what to do in crises [1, 14]. Prior work examined help-seeking and help-providing messages on social media and classified different categories of support that people seek or provide [21, 24]. Research also found potential problems in the supporting process, e.g., the help-seekers and help-providers were not well matched [31].

During the COVID-19 pandemic, Weibo—the most popular Chinese microblogging system—created an online community, named *COVID-19 Super-Topic*, specifically for users to seek or to provide help. Users can post to the *COVID-19 Super-Topic* about their (or family members' or friends') symptoms or medical issues [10] to call for help. Users can also comment on specific help-seeking posts to provide support. The super-topic community quickly became popular, receiving a lot of comments.

Although a rich body of research analyzed what people published on social media in crises, including help-seeking and help-providing messages [8, 19, 21], it is still not clear why and how people made comments on help-seeking posts to provide support. The COVID-19 pandemic is a special crisis in that people had limited knowledge about the virus and effective treatment at the beginning, thus it was challenging for people to offer help. There is a lack of investigation into people's commenting behaviors in this context. To address the research gap, we explored people's commenting behaviors with regard to the COVID-19 super-topic posts by conducting semi-structured interviews with 23 commenters. We focused on exploring why and how online users provided help through comments. We found that geographical proximity and expertise level affected users' commenting behaviors. For example, local people and experts tended to focus on detailed information and provide more detailed and practical information; whereas non-local or non-experts tended to help spread information through a "copy-and-paste" strategy and act as a bridge between help-seekers and influencers on social media. Some commenters also exhibited avoidance behaviors due

to their sense of powerlessness and frustration in this long-lasting pandemic. This study contributes new understandings of motivations and strategies behind online support on social media in crisis situations. Our findings also provide insights into the design of social media that can better support people to seek help from the general public during crises.

2 RELATED WORK

2.1 Social Media Use in Crises

Crises usually cause uncertainty, and people try to reduce that uncertainty [2] as well as improve self-efficacy [18] by obtaining relevant information. During crises, social media is viewed as a critical source of time-sensitive information [5]. Research in *crisis informatics* has explored social media use in a wide range of crisis events, including immediate crises, such as bombings [15], and lasting crises, such as epidemics [38, 42]. The findings showed that the use of social media in crises was affected by the period [42] and situation [12, 38] of the crises. Besides, as most crisis events are strongly regional, studies investigated the role of location in online behaviors and found that locals and non-locals focused on different kinds of information [33] and used them for different purposes [37]. Research further proposed the concept of emotional distance and found that it also played a critical role in information sharing and seeking behaviors during crises [15]. Construal-level theory shows that less spatial, social, temporal and hypothetical distance determines less psychological distance [39] and results in lower-level construal (a detailed, contextualized, and concrete manner) of items, which also suggests the potential effect of distance on online behaviors in crisis situations.

2.2 Help-Seeking and Help-Providing Behaviors on Social Media

It is well documented that help-seeking behaviors are common on social media during crises. People might turn to social media to understand the public information space and figure out what to do during unexpected crisis events [1, 4, 14, 32]. Some people tagged help-seeking posts on social media sites to make them more accessible in crises [28]. Researchers also considered help-providing messages on social media and examined different types of social support (emotional support, informational support, appraisal support, instrumental support) in crisis situations [21, 24]. One problem of the supporting process on social media is that help-seeking information and help-providing information are not well matched. Research suggested that help-seekers disseminated information about their needs without directly addressing particular groups or individuals [31]. To fix this problem, researchers proposed supportive technologies to identify help-seeking posts, recognize existing capabilities of the response communities, and provide resources to help-seekers [26, 27]. In the COVID-19 super-topic on Weibo, that problem might be solved since help-providers could directly comment on help-seeking posts to provide support.

2.3 Digital Volunteers in Crises

Considering that help providers play a similar role to “digital volunteer” in crises, we also turn to research on this topic. In unexpected

crisis events, official response organizations are sometimes not able to respond in typical fashion [13], so individuals would volunteer to provide rescue and relief efforts [9, 11, 17]. In crisis situations, digital volunteers play an important role in providing up-to-date information [35], verifying crisis data [22], helping affected people mobilize resources [34], and providing relief resources to affected people [30]. Research further classified digital volunteers into key-contributors who collected and processed much information and micro-contributors who performed small tasks for a voluntary response [25]. [16] proposed the role of moderators who regulated the requests and integrated information. Although digital volunteers respond quickly and play critical roles in helping others in need, their work does not always last. They are largely inactivated as social media and public attention to crisis events decrease [25], and they might abandon the identity of a digital volunteer because of tiredness or the fact that few responses are received [36]. In lasting crises, this decrease of attention may cause later help seekers not to receive as much help as former help seekers do.

2.4 Background: Sina Weibo and COVID-19 Super-Topic

Launched in 2009, Sina Weibo is now the most popular microblogging system in China. Weibo has a special feature called “super-topic” (“超级话题”) [41], which is similar to the hashtag on Twitter [7] in that it also allows users to tag a post so that it can be searched by the tag. Besides this function, every super-topic has its page and serves as a community for users with common interests to gather and discuss, which is similar to the “subreddit” feature on Reddit [23]. Given the spread of COVID-19, Weibo created a super-topic called “COVID-19 patient seek help” (“COVID-19”) on January 29th, 2020, for users to seek and provide help. Then a large number of users, both local and non-local, tried to support people in need by commenting on help-seeking posts on that super-topic.

Given prior literature and our research background, this study seeks to answer the research question: *Why and how do online users comment on help-seeking posts in the support community to provide help during the COVID-19 pandemic?*

3 METHODS

3.1 Participant Recruitment and Interview Protocol

To recruit participants from the COVID-19 super-topic, we first crawled all comments in it in April 21, 2020, and built a user pool including all 13,962 unique commenters who made a total of 24,414 comments. We randomly sampled 247 commenters and sent private messages to them. We received 34 initial responses and 23 participants finally consented to be interviewed. The 23 interviewees included six men and 17 women. Participants’ age ranged from 18 to 36 (mean = 24.4). We completed semi-structured interviews with all participants by voice and the same interview outline was used for them. Interviews lasted from 30 to 60 minutes (mean = 39.3), and each participant received 50 Yuan.

Our interview questions were designed according to individuals’ thoughts and actions during the whole process of commenting on

help-seeking posts. We first asked participants about their experiences and motivations for using Weibo and joining the COVID-19 super-topic. Then we talked about their considerations before commenting, the specific comments they made, and the reason for making these comments. Our interviews were not limited to the initial outline and emergent topics might be included, e.g., the length of attention initially. To avoid discomfort caused by interview questions, especially for those affected by COVID-19, participants could skip any questions or end the interview at any time. In our study, all participants finished the interview and answered all of the questions.

3.2 Data Analysis

We recorded all interviews with participants' consent and transcribed the audio data. Then we analyzed transcripts using grounded approach [6] and thematic analysis [3]. Specifically, we conducted a preliminary data analysis of the first few interviews. Two researchers separately open-coded the transcripts at the sentence level and then grouped the codes into themes using axial coding. Then they discussed and refined those codes and themes until they reached agreements in them. After setting the initial codebook, researchers coded the remaining data guided by it. When they found new codes or themes that were not included in the initial codebook, they added them to it. After coding 21 interviews, they were not able to find any new codes or themes from the data, which indicated the saturation of the theory [6]. In the end, the codebook contained 157 themes, e.g., "searching for help-seeking posts," "unfamiliarity with local situation," and "acting as what others do"; 14 sub-categories, e.g., "selectively verification," "experts' strategy," and "copy-and-paste strategy"; and 4 categories: "motivation," "verification process," "comment strategy," and "attention decay." The overall Cohen's Kappa was 91%, which is considered acceptable.

4 FINDINGS

In this section, we present our findings based on the categories and subcategories developed in our code book. In the remainder of this paper, "P" refers to the participant code.

4.1 Noticing COVID-19 Super-Topic on Weibo

4.1.1 Coming across Help-Seeking Posts. COVID-19 has become a hot topic in social media after the pandemic broke out, but help-seeking posts did not initially receive widespread attention. As described by participants in our study, they often experienced a process of coming across some clues of help-seeking posts, then actively looking for them and finally finding them. Some participants viewed Weibo as a major information source and first found the COVID-19 super-topic in "Weibo Trends" ("热搜榜", similar to "Twitter Trends") (P12) or from posts forwarded by influencers (P7, P10). Other participants first found out about the topic in other platforms where someone referred to touching stories (P9) or shared a screenshot (P14) from the COVID-19 super-topic. They then followed the clues and found it in Weibo.

4.1.2 Actively Looking for Help-Seeking Posts. Unlike those accidentally seeing the COVID-19 super-topic, some people directly

searched for it because of anxiety or willingness to help others. P1 said she lived far away from Wuhan (the most affected city in China) and searched this topic in the hope of supporting help-seekers: *"I guessed there might be some useful information that was not spread effectively enough and some people were not aware of it. I thought maybe I could help with it, so I searched for posts requesting help on Weibo. The COVID-19 super-topic appeared in the result list."* P18 explained that he was right in Wuhan then so he paid close attention to all COVID-related information and searched for as many relevant posts as he could. The COVID-19 super-topic was also in the search results. P21 and P23 did not have the habit of using Weibo, but they turned to Weibo to look for help-seeking posts because of its popularity. A medical professional (P21) described such shifts in the choice of social media in the crisis: *"I originally thought about posting some articles in my own account in Wechat to help people in need, but the number of its followers is quite small—only some of my friends followed that account, so I couldn't directly connect with help-seekers. I finally chose Weibo, a more popular public platform because I thought the information could spread more quickly to a wider range there."*

4.2 Before Taking Action: Verifying or not?

4.2.1 Selective Verification: Ineffectiveness. As hoaxes and deception are prevalent in social media [20], skepticism is often produced among users [44]. In our study, P4 expressed that it was inevitable that some people might take advantage of others' kindness, which might hinder the process of providing help. Therefore, it is not surprising that some participants (10 out of 23) claimed they had verified the original post at least once before deciding whether to comment or not. There are two basic ways to verify the authenticity of the help-seeking posts: to check users' profiles (adopted by 7 participants) and to check detailed information shared in the post (adopted by 5 participants). However, due to the time limit, participants might only skim and roughly check some posts rather than carefully verify each of them in detail. For example, P12 claimed that there were too many posts for her to verify them all, so she only checked the posts that impressed her with touching stories. P14 was similar to P12, but she had changed her verifying behavior over time. She tried to check each post initially, but then she noticed the ineffectiveness of doing that, and finally realized *"I would rather take time to comment on more posts than verify each of them."*

4.2.2 Skipping the Step of Verification: Needlessness and Impossibility. More than half of participants (13 out of 23) said they did not verify help-seeking posts before commenting at all, which is surprising in such a weak-tie social media platform [43]. P9 and P15 explained that the seeming richness of content and the disclosure of personal information made them feel that those posts were generally truthful, and P13 expressed that the similar format of the posts in the COVID-19 super-topic increased their credibility. P10 not only fundamentally denied the significance of verification but also questioned the method of verification: *"I didn't want to miss any opportunity to help. I hoped and tried to believe that everyone on that super-topic was really calling for help and nobody deceived others. After all, you don't have any practical way to verify whether it's true. Some people registered a new account only for*

posting help-seeking information, so there were no other posts in their account for verification.”

Taken as a whole, we found that when it came to health-related or even life-threatening help-seeking posts, people might only selectively verify them or even skip the whole verification process for three main reasons: needlessness, impossibility, or ineffectiveness.

4.3 Commenting Strategies

Whether or not participants went through the step of verification, they then started thinking about how to comment and offer help. By asking about the specific motivation and action when commenting, we found that participants with different geographic locations and expertise levels adopted different strategies.

4.3.1 Remote Users' Copy-and-Paste Strategy. Many participants (11 out of 23) explained that they perceived unfamiliarity and confusion caused by spatial distance and lack of medical knowledge. These people were prone to feelings of powerlessness and were thus eager to do something to relieve those feelings, but they felt difficult to find a good way. P15 shared that: *“For this kind of posts, I could not really help a lot. I was not in Wuhan, so I didn't know local situations; I was not a medical student, so I was not able to offer professional advice. For the help-seekers on Weibo, I really wanted to do something to help them, but I just didn't know where to start.”*

Therefore, to deal with feelings of powerlessness and depression, some participants adopted a special “copy-and-paste” strategy. In other words, they looked at what other people did and published similar comments to help others. P13 was a good example: *“I found that lots of comments in the COVID-19 super-topic mentioned multiple influencers. I didn't really know who they were, but since others all mentioned them, and it seemed if help-seekers contacted them, they could receive help more quickly, I just copied and content from other users, and then commented and mentioned the same influencer list. I did what I saw others did, and I hope it would work to help people.”* Besides individual activities, we also found cooperation between users who adopted the copy-and-paste strategy. Specifically, some users helped others to propagate more comprehensive and useful information. P9, who had adopted the copy-and-paste strategy alone at first, shared this kind of experience: *“Sometimes I would comment [on help-seeking posts] by simply copying and pasting what others did. Once an enthusiastic man sent me private messages to give me a comprehensive summary of information such as which phone number help-seekers could call, which influencer they could turn to for help, etc. He said that he had seen me post several simple comments and suggested me to copied and pasted a more comprehensive version.”*

When people employed the copy-and-paste strategy to mention a list of influencers, they hoped these potential help-providers could see their comments and really provide help. So they would feel happy and fulfilled if they received any feedback from the influencers. As P13 said, she was quite surprised to receive replies from an influencer. She mentioned: *“I was surprised when I mentioned an influencer, and he quickly replied to me, saying ‘leave it to me.’ I knew I couldn't actually provide much help, but when I got the reply from that influencer, I felt like that I've done something useful. Although I didn't know if he would really assist the help-seeker after saying those words, it is good to get the reply...It's kind of magical, surprising and reassuring as if what I did was quite meaningful.”* However, this

did not usually happen; for other participants who adopted the copy-and-paste strategy, they did not get any feedback or know if their comments worked. As P18 said, *“The final results of most help seeking posts are satisfactory, but it is not clear whether our comments made a difference.”* To find out what influencers thought of this strategy, we turned to an influencer (P21) for her opinion. In contrast to other participants' expectations, she explained that too many repeated mentions would actually reduce the efficiency of the problem-solving process: *“I saw so many commenters mentioning me. They seemed to post the same content over and over again. But to be honest, it was more efficient for me to directly look for people in need in the COVID-19 super-topic and reach out to them by myself, rather than following others' distributed mentions, because they were messy and repetitive.”*

To sum up, copy-and-paste is a relatively popular strategy that 6 out of 23 participants have adopted and 10 other participants have seen. For those who adopted a copy-and-paste strategy, they tended to publish highly similar, or even the same comments on different posts over and over again, while our interview with an influencer questions the effectiveness of this strategy.

4.3.2 Locals' and Experts' Targeted Commenting Strategies. As mentioned above, the copy-and-paste strategy is facilitated by spatial distance and lack of knowledge. Therefore, participants who did not face these problems took different commenting strategies. P20 lived in Wuhan and knew that city well, so he provided help according to the location the help-seeker reported rather than copying the same content: *“I live in Wuhan, and I remember that most of [help-seeking] posts were from Wuhan. I would definitely check if the help-seekers reported they were in Wuchang, Hanyang or Hankou (districts in Wuhan city), and then gave targeted information. It would be useless if someone was in Qiaokou but you gave him information about Wuchang or Qingshan. Non-local people might be unfamiliar with this detailed information in Wuhan, but as a local citizen, I am quite clear about it.”* Similar to P20, another local participant P19 also mentioned that she paid much attention to the location information in each post, and provided targeted help according to it.

In addition to local participants, participants with medical knowledge also provided targeted help. P11 was not local, but he had much medical knowledge and therefore complained about the high degree of similarity of comments. He noted that some people made several medical suggestions, but their comments on different posts were the same. He stressed that the treatment method should vary according to gender, age, constitution, and condition. That is why he tailored his comments according to the personal characteristics of different help-seekers instead of copying and pasting others' comments.

4.4 Attention Decay in Help-seeking Information

4.4.1 Deliberately Avoiding Looking at Help-seeking Posts. The COVID-19 pandemic has a relatively long duration, unlike widely studied instantaneous events such as shootings and earthquakes. Some participants (12 out of 23) in our study exhibited an avoidance behavior that they consciously stopped looking for help-seeking posts after a short period. P12 shared that she stopped following the update in the COVID-19 super-topic due to emotional factors:

“I spent a long time on them every day and took in too much negative information that I even lost my appetite. At that time, I even had a ridiculous thought that I couldn’t laugh. I reckoned that other people were in hot water, so I had no right to be happy since I could not provide any useful help. I realized that I had some mental health problems, so I forced myself to stop watching those posts and escape from that information.” P10 and P16 are slightly different from P12 in that their cessation is triggered by external conditions such as account abnormality, but the underlying cause was the same.

4.4.2 Lasting Attention due to Emotional Proximity and Expertise.

There also existed some participants (7 out of 23) who claimed that they kept paying attention to the COVID-19 super-topic. Some of them were local people or had strong emotional ties to Wuhan. P19 was in Wuhan then and shared the feelings of nervousness and worries. She was quite concerned about the situation in her neighborhood, and therefore *“clicked on the topic and looked through the posts there many times a day.”* While P22 was not in Wuhan then, she also had a strong emotional connection with Wuhan since she had lived and worked there for more than ten years and viewed Wuhan as her second hometown. She kept paying attention to the COVID-19 super-topic because she worried about her colleagues and friends who were in Wuhan then. Participant P21 didn’t stop for another reason: a professional reading style. Unlike those easily attracted to touching descriptions, P21 was less susceptible to traumatic words and therefore kept following help-seeking posts without being held back by negative feelings: *“Probably most people were attracted to traumatic words like ‘all my family except me are infected’ or young people saying ‘please save my parents’. But I paid little attention to these words because they were not informative for me to provide help. I focused on other truly important information such as help-seekers’ underlying illness. Therefore I was not affected by negative words and kept following the requests of help-seekers.”*

5 DISCUSSION

5.1 Distance and Commenting Patterns

This study shows that although social media broke down geographical barriers and reframes social networks, the spatial and social distance still affected users’ perception of help-seeking posts, their feelings, and following behaviors in the crisis support community. Our findings are aligned with construal-level theory [39], which claims that less spatial and social distances result in lower-level construal (a detailed, contextualized, and concrete manner) of items. We found that participants with more spatial proximity or medical knowledge (social proximity) focused more on detailed content in help-seeking posts (e.g., location and underlying illness) and formed low-level construal, while participants with greater spatial or social distance were easily attracted to emotion-related content and shaped high-level construal.

5.2 Influencers and the Copy-and-Paste Strategy

In this study, most participants acted as micro-contributors [25] who did not offer direct assistance but mentioned influencers in Weibo to call for help. Prior work found that help-seekers often

disseminated information about their needs without directly contacting help-providers in crisis situations [31], and researchers developed some methods to match help-providers and help-seekers [26, 27]. In this study, although the comment function made it possible for people to provide help directly to the help-seekers, most participants only mentioned a list of influencers because of geographic distance or lack of professional knowledge. These participants acted as a bridge between help-seekers and help-providers. We further found that these people adopted a particular “copy-and-paste” strategy where they looked at what other people did and copied their comments to then comment on more help-seeking posts. This strategy enabled individuals who could not provide direct help to contribute as much as they could and reduced the feelings of powerlessness and frustration. However, by interviewing an influencer mentioned by those users, we found that this strategy did not always work, and sometimes repetitive mentions could even reduce the efficiency of providing help.

5.3 Powerlessness and Avoidance Behavior

Though not included in the original interview scheme, avoidance behavior emerges as an essential topic in our interviews. Many participants mentioned that during such a lasting crisis event, they consciously stopped looking at help-seeking posts after a relatively short period, which was often less than a week. Avoidance behavior is similar to the performance of social media fatigue [29] in that they both lead to reduction of social media use. However, different from social media fatigue which is caused by overload, invasion of privacy, and invasion of work [40], avoidance behavior in this study is caused by powerlessness and distress after being exposed to too much negative information in a short period. It should be noted that not all participants exhibited this behavior. Our interviews show that professional knowledge, geographic and emotional connections made individuals keep paying attention to help-seeking posts.

5.4 Design Implications

This study indicates that comments under help-seeking posts were not equally valuable. Commenters who were closer to the epicenter of crises and those with medical expertise tended to offer targeted support rather than copy and paste the same content over and over again. Therefore, it may be beneficial for help-seekers if social media platforms could adjust sorting ways to make comments published by local people or relevant experts more visible in crisis support communities. Besides, this study suggests that many users acted as a bridge between influencers and help-seekers, but their messy and repeated behaviors might not really help. Thus, social media could identify the characteristics of help-seekers and automatically match them with potential help-providers to facilitate the problem-solving process. Additionally, since some people only paid attention to help-seeking posts for a short period, social media could dynamically increase the visibility of the entrance to support communities so that later posts could also receive timely support.

6 CONCLUSION

During the COVID-19 pandemic, Weibo established the COVID-19 super-topic where people could publish posts to seek help or

make comments on these posts to provide help. This study randomly selected and interviewed 23 commenters in that super-topic to examine why and how they commented on help-seeking posts to provide support in crises. Our findings suggest that locals and medical experts paid more attention to detailed information in help-seeking posts and provided targeted help, while non-locals and people lacking professional knowledge were easily attracted to touching stories and adopted a special “copy-and-paste” strategy. The latter commenters acted as a bridge between the help-seekers and influential help-providers on social media. However, the copy-and-paste strategy might not actually contribute to the higher efficiency of the help-providing process. We also found that non-locals and people lacking expertise might exhibit avoidance behaviors due to powerlessness and frustration in such an lasting crisis event. This study complements research on help-providing behaviors in crisis support communities and hopefully draws implications into how social media platforms can better support users in need during crises.

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